

Business Requirements Document (BRD)

Online Ringi System

(ORS)



Version: 4.1
Date: 13th February 2026

Contents

- **Document Control** **2**
  1. Version Control 2
  2. Stakeholders 2
- **Executive Summary** **4**
- **As-Is Process (Current State)** **4**
- **To-Be Process (Proposed State)** **5**
- **Business Objectives** **5**
- **Functional Requirements** **6**
- **Non-Functional Requirements** **28**
- **Process Flow** **30**
- **Hardware and Software Requirements** **34**
- **Change Request (CR) Management** **35**
  1. Change Request (CR) 35
  2. CR Management Approach 36
  3. Definition of Change Request 36
- **Dependencies, Assumptions, Out of Current Scope** **38**
  A. Dependencies on the Global Automotive Manufacturer 38
  B. Assumptions 38
  C. Out of Current Scope 39
- **Project Communication and Control** **40**
  A. Documentation 40
  B. Governance 41
- **Commercial Proposal & Terms & Conditions** **42**
  - Commercials for Technology Services Provider Development Services for Application 42
  - Terms and Conditions 42
- **Technology Services Provider Business Team Members & Escalation Metrics** **43**
  1. Escalation Matrix for Application Support Project 43
  2. Business Team 43
- **Document Acceptance Sign-Off** **43**

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1. Version Control

Date	Document Version Number	Author	Verified By
8th Oct 2026	v4.1	Technology Services Provider Solution Team	Technology Services Provider Business Team

2. Stakeholders

Organization	Designation / Dept	Name	Email ID

Executive Summary

Global Automotive Manufacturer currently operates a legacy, Ringi system that supports financial and informational approval workflows. The existing system, developed approximately 8-10 years ago, has limitations in scalability, configurability, mobility, audit traceability and real-time integration with SAP. These constraints impact approval turnaround time, budget governance, audit readiness, and overall operational efficiency.

The proposed Online Ringi System (ORS) aims to replace the legacy application with a modern, secure, and configurable web-based platform. The new system will enable dynamic approval workflows, real-time SAP budget validation and consumption, robust audit trails, and role-based dashboards. It will support both Financial and Information Ringis, with the ability to link them seamlessly to ensure transparency and contextual decision-making.

The solution will be developed using a modern technology stack and deployed on Global Automotive Manufacturer's on-premise infrastructure. It will integrate with SAP through approved APIs for budget validation, consumption, and reversal, while ensuring clear separation of responsibilities between the ORS and SAP systems. Mobile-responsive access, automated notifications, configurable reporting and centralized administration will further enhance usability and governance.

The system shall support at least 1,500 users initially, scale to 2,500 users within one year, and handle an annual volume of approximately 5,000 Ringis.

This BRD documents the business requirements, functional scope, non-functional expectations, assumptions, and dependencies for the Online Ringi System. Upon sign-off, this document will serve as the baseline for solution design, development, testing, and delivery. Any changes beyond this scope will be managed through the formal Change Request (CR) process.

As-Is Process (Current State)

- Global Automotive Manufacturer currently uses a legacy Ringi system developed several years ago.
- Ringi creation and approval workflows are largely rigid and require manual intervention for exceptions.
- Approval hierarchies are hard-coded and difficult to modify when organizational changes occur.
- Budget validation against SAP is not fully real-time and often requires manual verification by Finance teams.
- Cancellation and reassignment of Ringis involve manual coordination through emails and follow-ups.
- Limited visibility exists for tracking pending, overdue, or bottleneck approvals across departments and plants.
- Reporting and dashboards are basic, with limited filtering and audit-ready insights.
- Audit trails, remarks, and document traceability are fragmented and not easily accessible in a consolidated manner.
- The system offers limited mobile accessibility and lacks a modern user experience.
- Manual follow-ups are required to ensure timely approvals, leading to delays and inefficiencies.

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To-Be Process (Proposed State)

- A centralized, web-based Online Ringi System will replace the legacy application.
- Users will create Financial and Information Ringis through a structured, intuitive interface.
- Approval workflows will be dynamically configured based on department, Ringi type, amount, and financial year.
- Near real-time SAP integration via intermediary tables with updated budget codes will validate budget availability during Ringi creation and approval. The new Ringi system will perform data import and export through scheduled batch processes executed at defined time intervals.
- Automated routing will ensure Ringis move seamlessly across approval levels without manual intervention.
- Reassignment and transfer of Ringis will be system-driven with mandatory reason capture and audit logging.
- Financial Ringi cancellations will automatically trigger budget reversals via the new Ringi system in the intermediary table which will be linked with SAP.
- Role-based dashboards will provide real-time visibility of Ringi status, pending actions, and overdue approvals.
- Automated notifications and reminders will reduce dependency on manual follow-ups.
- A consolidated Ringi PDF, including full approval history and all supporting documents, will be automatically generated for audit purposes. Once approved, the final Ringi PDF will be emailed to the initiator.
- Ringi records generated in the legacy system will be displayed in a separate window with view-only access (no create, edit, or delete).
- The system will support mobile access and enhanced reporting and action for management and audit teams.
- The Ringi initiator will be able to pull back a Ringi after submission. Once pulled back, approvers will not be able to approve it, only rejection will be allowed.

Business Objectives

The primary business objective of this project is to enhance the processes:

1. Digitize end-to-end Ringi lifecycle to eliminate manual and email-based approvals. Reduce approval cycle time through automated routing and reminders
2. Enable dynamic, rule-based approval workflows configurable without code changes.
3. Ensure continuity of approvals through reassignment and transfer mechanisms during employee absence or exit.
4. Enforce real-time budget validation against SAP during Financial Ringi creation.
5. Prevent budget overruns by blocking approvals beyond available budget.
6. Provide complete visibility of budget utilization, reversals, budget additions and balance at cost-center level.
7. Maintain a complete audit trail of all Ringi actions including approvals, rejections, holds, cancellations, and reassignments.
8. Provide centralized dashboards for real-time visibility of all Ringis across departments.
9. Deliver a modern, intuitive, and responsive web interface.
10. Reduce dependency on manual follow-ups through automated notifications.
11. Support multilingual usability (English and Japanese) to cater to diverse user groups.
12. Support configurable approval levels and workflow expansion.
13. The system shall allow users to search and filter Ringis by approval level (e.g., L1, L2, etc.)

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and Ringi type (Capital or Revenue), Ringi Number, Ringi Subject.

14. Division heads will have access to the Ringis raised by their own departments.
15. The system shall allow auto mapping of Ringis raised by users of a department to the respective Department Head/Division Head/Operations Head.
16. Enable automatic skipping and reassignment of Ringis based on user unavailability schedules.

Functional Requirements

The functional requirements define the system capabilities required to support Global Automotive Manufacturer's Ringi approval processes.

1. Ringi Creation & Initiation

#	Requirement Description
RCI-01	The system shall allow authorized users to create two types of Ringis: ● Financial Ringi - Used for approvals involving financial implications such as budgets or expenses. ● Information (Non-Financial) Ringi - Used for information sharing, notifications or approvals that do not involve any financial impact. Budget Transfer Ringi to come under Information Ringi. ● Special Class Ringi - Used for budgets greater than INR 1 crore and shall be created only by the Division Head & above and routed exclusively to the Managing Director (MD) for approval.
RCI-02	Users shall be able to create, edit, save as draft, submit, hold, pull-back or reject a Ringi based on their assigned role and the current status of the Ringi.
RCI-03	While creating a Ringi, users shall be required to select the following mandatory details: ● Plant / Location (Gurgaon, Kharkhoda, Warehouse, Regional Offices, GBP Office, Global Automotive Manufacturer All - manual) ● Department Code (autofill) ● Cost Center (autofill) ● Financial Year (manual) ● Dynamic Approval Matrix (based on Information & Financial Ringi) For Financial Ringis, the following additional fields shall be mandatory: ● GL Code ● Budget Type (Capital/Revenue) ● Cost Center
RCI-04	Before a Financial Ringi is created, the system shall check whether sufficient budget is available for the selected department. ● If sufficient budget is available, the required amount shall be temporarily reserved for the Ringi until it completes the approval process. ● Upon final approval, the reserved amount shall be fully allocated to the Ringi. ● The reserved amount shall remain allocated to the same Ringi, even if the Ringi is sent back for correction or rework. ● If the Ringi is rejected by every approver and the initiator , the reserved amount shall be released and added back to the available budget. For an Information Ringi, the budget amount field shall also be available as optional and captured as

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	part of the Ringi details, even though it does not involve financial approval and validation.
RCI-05	The system shall automatically capture and store the following details at the time of Ringi creation: ● Ringi creation date and time ● Name of the Ringi creator ● Employee ID of the Ringi creator ● Designation of the Ringi creator ● Department of the Ringi creator ● Ringi Number (draft/actual) These details will be system-generated and not editable by the user.
RCI-06	For Financial Ringis, the amount field shall be mandatory and must be linked to a single budget code only.
RCI-07	While creating a Ringi, the user shall be able to select the required approval level. The system will route the Ringi for approvals up to the selected level only.
RCI-08	The subject title of the Ringi shall be a mandatory field with a maximum limit of 150 characters. This title will be used as a reference for searching and identifying Ringis within the system.
RCI-09	The Ringi content section shall include a built-in text editor that allows users to format content as required. The editor shall support the following features: ● Text formatting: bold, italics, underline ● Text styles: heading levels (H1–H4), title, subtitle ● Font options: size and color ● Text highlighting ● Hyperlink insertion ● Image insertion ● Text alignment (left, center, right, justified) ● Bullet points and numbered lists ● Indentation and spacing controls The text editor shall provide the ability for users to resize tables by adjusting individual column widths and row heights, as well as resizing the overall table dimensions, using a drag-and-drop interaction. The Ringi content body shall also support a 5W-1H (What, Why, Who, When, Where, How) explanation in a tabular format, which can be optionally filled by the user.
RCI-10	The system shall support the creation of Joint Initiators for a Ringi. Users shall be able to dynamically add one or more initiating authorities while creating the Ringi. Joint Initiators may belong to the same department or different departments, as per the requirement.
RCI-11	The system shall synchronize Employee Master data with the existing HRMS used at Global Automotive Manufacturer, following the table structure defined in the HRMS. The system shall use employee attributes provided by Global Automotive Manufacturer including Employee Name, Employee ID, Email ID, Designation, Department etc. for system configuration and user management.
RCI-12	The default approval workflow for a Ringi shall follow the below sequence:

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	• Joint Initiating Authority (JIA) • Department Approver • Other Department Approver • Finance Approver • L7 Approver (Below Section Head) • L6 Approver (Section Head / Section Manager) • L5 Approver (Department Head / Division Manager) • L4 Approver (Division Head / Division Manager) • L3 Approver (Operation Head / Operation Manager) • L2 Approver (Director) • L1 Approver (MD) The Ringi shall move sequentially through each approval level based on the defined workflow until final approval is completed. The workflow to be configured basis on Operations. For an Information Ringi, the Finance Approver shall be optional. For a Financial Ringi, the Finance Approver shall be mandatory.
RCI-13	Users shall be able to attach supporting documents during Ringi creation. The system shall support following files to be uploaded - jpg, png, jpeg, pdfs, docs, ppt, spreadsheet, txt, outlook email file. If applicable, all hyperlinks embedded within the Outlook email file shall be disabled by the system.
RCI-14	While creating a Financial Ringi, users shall be able to link it to an existing Information Ringi and vice versa.
RCI-15	The details of the linked Ringi shall be viewable on the same screen without requiring navigation to a separate browser tab.
RCI-16	If a Ringi is saved as a draft, it shall be visible only to the creator of the Ringi. Until the Ringi is submitted, it shall carry a temporary(draft) Ringi number. Once the Ringi is submitted, the system shall automatically generate a unique Ringi number based on the plant-wise sequencing logic provided by Global Automotive Manufacturer. Once generated, a Ringi number shall be reserved for that specific Ringi and shall not be reused for any other Ringi.
RCI-17	Once a Ringi is submitted or approved, the user won't be able to make any changes in the existing Ringi.
RCI-18	Once a Ringi is approved, the next approver will have the options to approve, reject, hold the Ringi. The system shall require the approver to enter remarks before performing any action. The remarks field shall enforce a minimum of 2 characters and a maximum of 450 characters.
RCI-19	In case of Ringi Rejection for ringi update, the Ringi will have to go through iteration at each level of users involved and rejected from their account as well before it becomes editable by the Ringi creator and any changes can be made to it. The system shall allow editing of a Ringi only by the Initiator, JIA and the originating department's approver. Once a Ringi is submitted by the Initiator or JIA, no further updates shall be permitted by them unless the Ringi is rejected by the originating department's approver. Once the Ringi content is updated, the system shall send notifications to all previous approvers in the approval workflow.
RCI-20	When a Ringi is rejected for revision, the Ringi creator shall be allowed to update the Ringi as required. This includes:

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	• Adding or modifying supporting documents • Adding new approvers or stakeholders • Updating existing details as part of resubmission If any existing stakeholders are removed during the revision, the system shall notify them through system notifications and email alerts.
RCI-21	The system shall maintain a complete audit trail of all approval stages, including the remarks entered, actions taken and the corresponding date and time for each action.
RCI-22	Upon final approval, the system shall auto-generate a consolidated Ringi PDF including: • Ringi details • Approval history • Remarks • Attached documents Technology Services Provider shall design an improved Ringi format to enhance clarity and readability. Once a Ringi is approved by the final approver, the system shall send an email notification to the Initiator with the approved Ringi attached as a restricted PDF. The attached Ringi PDF shall be access-restricted to prevent unauthorized viewing, editing, or sharing. The Ringi PDF document shall display the message “Confidential Document - Global Automotive Manufacturer Internal Use Only” at the bottom of every page.
RCI-23	At every actionable stage, from Ringi creation through final approval, the system shall automatically trigger email notifications to all relevant stakeholders involved in the Ringi process.
RCI-24	If a Ringi is on hold or remains pending at any approval stage beyond the defined time limit, the system shall automatically trigger reminder emails to the concerned approver. • For Normal Ringis, the first reminder shall be sent after 3 days of inactivity, followed by subsequent reminders every 24 hours until action is taken. • For S-Class (Special Class) Ringis, reminder emails shall be triggered after 1 day of inactivity and continue at defined intervals until the Ringi is acted upon. The system shall send only one email notification per user per day for all pending actions, including Ringis awaiting approval, rejection, or currently on hold.
RCI-25	When a Ringi is exported, all supporting documents attached to the Ringi will be downloaded together as part of the export in a zip folder.
RCI-26	The system shall provide the Ringi Initiator with an additional option to Pull Back a submitted Ringi. In case of a Pull Back Ringi request, the system shall follow the same approval hierarchy as the original Ringi workflow. Each approver in the hierarchy shall be required to accept the pull back request before it is routed back to the Ringi initiator.
RCI-27	In the case of a Budget Transfer Ringi, the requesting department shall create the Ringi and specify the source department’s cost center from which the budget will be transferred. The cost center field shall be a numeric, user-entered field and shall not be provided as a dropdown or selectable list.

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	The system shall validate the entered cost center and ensure that sufficient available budget exists before allowing the Budget Transfer Ringi to proceed.
RCI-28	Once a Budget Transfer Ringi is submitted and approved by the respective Department Head or Other Department Head, the system shall route it to the Costing Team for budget verification before proceeding further in the approval hierarchy.
RCI-29	If a Ringi approver is assigned at multiple approval levels, the system shall require only one approval from that user and automatically mark the remaining levels as approved in which they are involved.
RCI-30	The system shall ensure that the JIA role is not assigned below the Section Head level. If a Section Head is not appointed, the system shall allow the Initiator to be below Assistant Manager level for the JIA role.
RCI-31	For a Financial Ringi, the system shall reserve the budget amount at the time of creation, including when the Ringi is saved as a draft. The system shall ensure that only the remaining available (unreserved) budget is visible and usable for new Ringi initiations. The budget shall be formally consumed only after the Ringi receives final approval. When Ringi is rejected, the budget will be released when final rejection is done from the initiator.
RCI-32	An Information Ringi shall be categorized into two types: Financial Impact and Non-Financial Impact. If the Financial Impact option is selected, the inclusion of Finance Approvers in the Ringi shall be mandatory. In the case of a Non-Financial Impact Information Ringi, selection of Finance Approvers shall not be required.
RCI-33	The selection of whether a Ringi is classified as S-Class or Non S-Class(Regular) Ringi shall be applicable before the selection of Financial or Information Ringi Type.
RCI-34	Users at the level of Division Head and above shall have access to create S-Class Ringis. The approval workflow for an S-Class Ringi shall follow the sequence: Division Head → Division Manager → Operation Head / Executive Officer → Operation Manager / Executive Officer → Finance → MDO → CFO → Director 1 → Director 2 → MD.
RCI-35	For all L1 and L2 Ringi, review by Finance Approver 1 shall be mandatory.
RCI-36	The ability to put a Ringi on hold, add comments and upload additional attachments shall be available to Division Head and above.
RCI-37	The system shall allow blocking specific Ringi types (Revenue/Capital) for selected departments or divisions through Admin.

2. Approval Workflow Management

#	Requirement Description
AWM-01	The system shall support configurable dynamic approval workflows.

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AWM-02	The Admin user shall be able to configure amount-based conditions for different types of Ringis, such as Capital and Revenue Ringis for each department. Based on the configured amount thresholds, the system shall determine the applicable approval level for each Ringi. The actual approver at each level shall be selected by the Ringi creator, while the system will control and restrict the approval levels as per the configuration defined by the Admin user. All workflow and approval configurations shall be defined operation-wise and inherited by the departments mapped to that operation. The system shall support department mapping to an Operation.
AWM-03	The system shall support configurable and dynamic approval workflows, along with the default MD, Director, Finance Approvers, Department Approvers, Other Department Approvers and JIA Approvers as per the operations policy. Currently, only the Purchase Policy is available. However, the system shall be designed to support policies for other operations, which shall be configured and managed by Admin users.
AWM-04	For each approval level, the system shall allow the Ringi creator to add additional approvers beyond the default configuration. By default, the system shall require at least one approver per approval level. The Ringi creator shall be able to optionally add JIA and other department approvers at any approval level. Approval from the creator's own Department Head shall be mandatory and cannot be removed.
AWM-05	Once submitted, the Ringi shall be auto-routed to approvers as per configured rules.
AWM-06	Approvers shall be able to perform the following actions: ● Approve ● Reject (Document Required, Modification in Ringi and Complete Ringi Reject) ● Hold (Also Resume)
AWM-07	The system shall require users to enter mandatory remarks at every approval action and at each approval stage before proceeding further in the workflow.
AWM-08	All remarks shall be logged and visible to all stakeholders involved in the Ringi.
AWM-09	The rules for Approval Workflow Management can only be configured by the Admin user.
AWM-10	The system shall allow the Admin to configure the number of approvers required at each approval level. The Admin shall define a minimum number of approvers for each level. The Ringi creator shall be allowed to add additional approvers beyond the minimum configured at any level.

3. Ringi Reassignment & Transfer

#	Requirement Description

RRT-01	The system shall allow reassignment of Ringis in case of approver unavailability or employee exit.
RRT-02	Reassignment shall require mandatory reason capture.
RRT-03	Ringi Initiators shall not have the ability to trigger reassignment. Only the immediate higher-level (+1) approver shall be authorized to reassign the Ringi to another user at the same approval level.
RRT-04	Admin also shall have the ability to transfer Ringis between users during organizational changes.
RRT-05	All reassignment actions shall be audit logged.
RRT-06	If an old employee leaves, ownership of pending Ringis can be transferred to a new employee. No ownership change can be done for the approved Ringis. Only the Admin user will have the right to transfer Ringis from one user to another.
RRT-07	The system shall support the transfer of approval responsibility from one approver to another in cases such as employee replacement, role change or unavailability of an approver. All such changes shall be fully tracked by the system and appropriate system notifications and email alerts shall be triggered to all stakeholders who have already taken action on the Ringi.
RRT-08	The system shall allow the next higher approval authority (+1) to reassign a Ringi when the designated approver is unavailable and shall record the user who performed the reassignment in the audit trail. The new reassigned user will then have the authority to perform action on Ringi.
RRT-09	Users at division head level and above have access to all Ringis raised for their division and operation.

4. Ringi Access and Role Mapping

#	Requirement Description
RARM-01	The system shall allow Heads and Managers to view and access all Ringis raised by users within their assigned section, department, division, or operation.
RARM-02	Users shall be able to view only the Ringis they have created, unless they are assigned as the Head or Manager of the respective organizational unit.
RARM-03	Users shall not be permitted to view Ringis raised by users from other sections, departments, divisions, or operations outside their scope of authority.
RARM-04	Access to Ringis shall be determined by the user's current role and organizational mapping, not by individual user-to-user sharing.
RARM-05	When a Head or Manager role is transferred to another user: • The newly assigned user shall automatically gain access to all Ringis associated with that role. • Users shall be able to view all Ringis they have raised and those in which they are involved as an approver, across both in-progress and approved statuses. • The previous role holder shall lose access to the new Ringis from their previous Dept/Division/Operation.

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RARM-06	The admin user will have the right to manually transfer ownership and accountability of Ringis during role changes.
RARM-07	Hierarchy currently/to be followed in the system: Section Head/Manager < Dept Head/Manager < Division Head/Manager < Operation Head/Manager < Executive Officer < Director < Managing Director
RARM-08	For users managing multiple organizational units limited to head/manager, the system shall provide a role-based mapping that links the user to all relevant departments, divisions, or operations.

5. Ringi Cancellation Management

#	Requirement Description
RCM-01	Information Ringis shall be cancellable without budget impact.
RCM-02	Financial Ringis shall be cancellable only if: ● No Purchase Order (PO) is generated, or ● Associated PO has been cancelled
RCM-03	The system shall allow cancellation of an approved Financial Ringi, even after PO generation, only if the PO Used Amount is zero and the PO Balance Amount is equal to the Ringi Amount. The system shall allow an Information Ringi to be linked before initiating the cancellation of a Financial Ringi. The cancellation Information Ringi to follow the same level of approval which it followed at the time of approval for the Financial Ringi. One same cancellation Information Ringi can be used for linking with multiple Financial Ringi cancellation.
RCM-04	Cancellation shall require acknowledgement from all stakeholders involved in the Ringi lifecycle.
RCM-05	Upon cancellation of a Financial Ringi, the system shall automatically reverse the consumed budget in SAP.
RCM-06	All cancellation actions shall be logged with user, timestamp and reason.
RCM-07	Automated notifications shall be sent to all stakeholders upon cancellation.
RCM-08	If a Ringi is rejected (complete Ringi rejection) by a higher authority, it cannot be re-initiated. Only Ringis that are rejected for modification in content or more documents required can be revised and re-initiated by the initiator. For a rejected Ringi, the allocated budget shall not be released automatically, it will require rejection from all approver hierarchy before the budget is released back to the available pool.
RCM-09	Only the Admin user will have the permission to cancel an approved Ringi directly from the system and system notifications and emails to be triggered to all stakeholders involved for the approval of that Ringi. An Information Ringi can be linked with the cancellation of an approved Ringi before the action is taken by the admin.
RCM-10	The system shall not allow any approved or rejected Ringi to be deleted or hidden under any circumstances, including by system administrators. Once a Ringi number is generated, the

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	corresponding Ringi shall remain permanently viewable in the system.
RCM-11	For the interim phase, the system shall allow Admin users to cancel Financial Ringis until SAP integration is successfully developed and implemented and made available by the Global Automotive Manufacturer vendor, the system shall permit only Admin users to cancel Financial Ringis.

6. Ringi Edit & Update

#	Requirement Description
REU-01	The system shall allow editing of a Ringi only by the following roles: • Ringi Initiator • Joint Initiating Authority (JIA) • Initiator's own Department Head No other users shall have permission to edit the Ringi content.
REU-02	A Ringi can be edited only until it is submitted for approval by the Initiator's own Department Head.
REU-03	Once submitted, the Ringi shall become read-only. Editing will be re-enabled only if the Ringi is rejected by any approver (e.g., JIA, Department Approver, Other Department Approver, Finance Approver).
REU-04	A Ringi must be rejected at each approval level before it returns to the Initiator, and it will become editable only after all required rejections are completed.
REU-05	The Ringi History/Audit Log shall capture: • User who performed the action • Action taken (Submit, Approve, Reject, Edit, etc.) • Date and time of the action The history will not store document content changes, it will record who acted, what action was taken, and when.
REU-06	Once the Ringi content is updated by the JIA or Department Head, all previous approvers, including the Initiator, shall receive a notification.

7. Ringi Pullback

#	Requirement Description
RP-01	Only the Ringi Initiator shall be allowed to perform the Pull-Back action.
RP-02	The Pull-Back option shall be available only after submission and before final approval.
RP-03	Other roles (JIA, Department, Finance, Audit, etc.) shall not have permission to pull back a Ringi.
RP-04	When the Initiator pulls back a Ringi: • The Ringi shall be removed from the active approval workflow.

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	• All approval actions shall be disabled for approvers. • Approvers shall not be able to approve the Ringi, only a Reject action shall remain available.
RP-05	The system shall prevent any further approval once the Pull-Back is triggered.
RP-06	A pulled-back Ringi shall not become editable immediately. The Ringi shall return to the Initiator in editable mode only after it has been rejected at every applicable approval level. Until the full rejection cycle is completed, the Ringi shall remain read-only for the Initiator.
RP-07	On Pull-Back, the system shall update the Ringi status to reflect a “Pulled Back / Recall in Progress” state (or equivalent).
RP-08	Once all required rejections are recorded, the system shall enable edits on that Ringi. The system shall ensure the Ringi follows the defined approval hierarchy even during the rejection cycle.
RP-09	Every Pull-Back action shall be recorded in the Ringi History with: • User name (Initiator) • Action taken (Pull Back) • Date and time
RP-10	The system shall not allow Pull-Back after final approval of the Ringi.
RP-11	The system shall not provide any undo or restore option once a Ringi has been pulled back. After a Ringi is pulled back, the Initiator shall not be able to cancel the operation.
RP-12	In the event of a Ringi being pulled back or rejected, the system shall allow the user to modify only the title, content, and approver list, and shall restrict any changes to budget-related data.

8. Archived Ringi

#	Requirement Description
AR-01	The system shall display Ringi records generated in the legacy/old system as Archived Ringis.
AR-02	Archived Ringis shall be clearly identified as “Archived” or “Legacy” records.
AR-03	Archived Ringis shall not be mixed with active Ringis in standard workflows.
AR-04	Archived Ringis shall be available in read-only mode only.
AR-05	Archived Ringis will be accessible by all users of ORS.
AR-06	Archived Ringis shall be displayed in a separate page/window/section from active Ringis.
AR-07	Users shall be able to open and view: • Ringi details • Approval history (if available from the legacy system) • Attached documents (if migrated or linked)
AR-08	Users shall be able to search archived Ringis using key identifiers such as: • Ringi number • Subject / title

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	• Initiator name • Date range The system shall apply filters based on the logic defined in the new system, if applicable.
AR-09	Archived Ringis may be viewed or downloaded - no modifications to documents, attachments or metadata shall be permitted.
AR-10	No business actions shall be allowed on archived Ringis. Archived Ringis shall not: • Enter any approval workflow • Be pulled back, rejected, or resubmitted • Trigger notifications or emails
AR-11	The table structure and data dump to be provided by Global Automotive Manufacturer which will be stored in the new system.

9. Unavailability Schedule Configuration

#	Requirement Description
USC-01	The system shall allow users to define an Unavailability Schedule by specifying a Start Date and End Date during which they will be unavailable to act on Ringis.
USC-02	While configuring the unavailability schedule, the user shall be required to select an Alternate User to whom their Ringi responsibilities will be temporarily transferred.
USC-03	The system shall validate that: • The Start Date is less than or equal to the End Date. • The Alternate User is an active system user. • The Alternate User belongs to the same or permissible organizational scope, as defined by admin configuration.
USC-04	The unavailability schedule shall be editable or cancellable by the user only before the Start Date. Once the Start Date is reached, the schedule shall become read-only.
USC-05	During the active unavailability period, the system shall automatically skip the unavailable user in all applicable Ringi workflows.
USC-06	Any Ringi that: • Is newly assigned to the unavailable user during the unavailability period, or • Is already pending with the unavailable user at the start of the unavailability period shall be automatically transferred to the selected Alternate User.
USC-07	The Alternate User shall receive the Ringis with the same approval level, authority, and action rights as originally assigned to the unavailable user.
USC-08	The system shall ensure that the unavailable user: • Does not receive approval notifications during the unavailability period. • Cannot take any approval action on Ringis assigned during that period.

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10. New Budget Allocation

#	Requirement Description
NBA-01	The ORS application shall support tracking of new budget procurement created during the financial year.
NBA-02	The system shall address the current gap where SAP provides budget only at initial level, but does not track new or additional budgets allocated mid-year.
NBA-03	The system shall automatically record new budget allocations in ORS upon approval of a Financial Ringi that results in a budget credit. The system shall also provide an option to credit (add back) the budget amount to the corresponding Budget Code.
NBA-04	At the time of Ringi creation, the system shall display budget details in a tabular view, including: • SAP-provided budget for the selected Cost Center + GL Code • New budget allocations recorded in ORS for the same mapping
NBA-05	The system shall show: • Total Budget • Original Budget • Additional Budget (procured during the FY) • Utilized budget (during FY) • Unutilised balance amount
NBA-06	The users shall be able to view the budget table in a separate window, filtered based on the budget codes mapped to their respective department, division, or operation. For Division Head and Operational Head users, the system shall display separate budget tables corresponding to the budgets allocated to their respective scope.

11. All Ringi Directory

#	Requirement Description
ARD-01	All approved Ringis shall be searchable and accessible for viewing within the system for future reference.
ARD-02	All cancelled Ringis shall be clearly marked with a “Cancelled” status for easy identification. The Admin shall have the option to archive cancelled Ringis, after which they will no longer appear in search results or listings within the system.
ARD-03	The subject title of the Ringi shall be used as a key reference for searching and identifying Ringis within the system. Ringi can also be searched on the basis of filters like Ringi Type, Level, Financial Year, Budget Code etc.
ARD-04	All approved Ringis shall be visible to all users who have access to the new ORS application.

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ARD-05	All supporting documents, approver details, and the remarks captured at each approval stage shall be available for viewing along with the Ringi details.
ARD-06	A Ringi shall only be available for download in PDF format, along with all its attached supporting documents. No other actions (such as editing or modification) shall be permitted on the downloaded Ringi. All supporting documents related to that Ringi to be downloaded in zip file at once.
ARD-07	In case of Dept/Division change of a user, they shall be able to view Approved Ringis in which they were involved as an initiator or approver.

12. Policy Directory

#	Requirement Description
PD-01	The system shall include a Policy Directory module to centrally store and publish policy documents for the entire organization.
PD-02	The module shall serve as a single source of truth for all official policy documents.
PD-03	Only authorized Admin users shall be able to manage policy documents.
PD-04	Admin users shall be able to: • Upload new policy documents • Update or replace existing documents • Change document visibility (e.g., visible to all users or restricted roles) • Remove or archive outdated policies
PD-05	End users shall have view-only access to policy documents based on the assigned visibility rules.
PD-06	The system shall restrict file uploads to PDF format only. Upload of other file formats (e.g., .doc, .docx, .xls) shall not be permitted.
PD-07	The system shall allow users to: • Browse policies by category or type (if configured) • Search policies by name or keywords • View and download available policy PDFs
PD-08	All document management actions (upload, update, visibility change, delete/archive) shall be logged with user, action, and timestamp for audit purposes.

13. Mobile Access for Ringi Action

#	Requirement Description
MARA-01	For specific Approver users, the new ORS application shall be accessible on mobile devices. These users shall be able to view, approve, or reject Ringis through the mobile interface.
MARA-02	These specific users will be able to access the ORS application via their personal mobile phone

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	outside of the Global Automotive Manufacturer network.
MARA-03	The application will allow users to view and approve/reject/hold requests via mobile devices. All other functionalities will be available only through the web interface.
MARA-04	The final discussion regarding providing mobile access to users is currently pending between Global Automotive Manufacturer and Technology Services Provider. The scope will be defined and documented once it is finalized.

14. Budget Management

#	Requirement Description
BM-01	The system shall validate budget availability in near real time from an intermediary table which is in-sync SAP during Financial Ringi submission.
BM-02	The system shall block ringi submission and approval if sufficient budget is not available.
BM-03	The system shall support budget transfer between departments through a Budget Transfer Ringi.
BM-04	The system shall allow users to create Budget Transfer Ringis for transferring surplus budget within an operation. Requestor users shall be permitted to initiate a Budget Transfer Ringi for transferring budget from a cost center to their own cost center and not vice-versa. The existing Ringi workflow, approval hierarchy and format shall remain unchanged for Budget Transfer Ringis, but fields like Budget Transfer From, Budget Transfer To, Budget Amount to Transfer. The budget transfer action shall be executed in SAP through the intermediary table. Technology Services Provider will update the intermediary table with the required budget transfer details, while the actual budget transfer processing in SAP shall be handled by Global Automotive Manufacturer and their SAP vendor. As per the Ringi policy, budget transfers within the same Operation, irrespective of the amount, shall require approval from L3 only. However, if the budget transfer is between two different Operations and the amount exceeds INR 1 million, approval from L1, in addition to L3, shall be required.
BM-05	Budget Transfer Ringis shall auto-assign the Costing Team for review and action.
BM-06	All budget movements shall be traceable and auditable.
BM-07	For Capex or Opex Ringi, before reaching Finance Approver-1 (FA1), the Ringi shall be routed through an additional approval level configurable by the Global Automotive Manufacturer Admin directly in the application. Users at this level shall be able to provide comments before the Ringi is forwarded to the FA1 approver.

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15. SAP Integration via Intermediary Table

#	Requirement Description
SIIT-01	The system shall integrate with an intermediary database/table provided by Global Automotive Manufacturer, which will remain in sync with the SAP system. The responsibility of maintaining synchronization between the intermediary tables and SAP shall lie with Global Automotive Manufacturer and its SAP vendor. The new system developed by Technology Services Provider will only read, write, and update data in the intermediary database as per the available records. All system operations will rely on the data present in this intermediary layer.
SIIT-02	During Ringi processing, the system shall fetch the approved budget amounts for each department from the intermediary database table.
SIIT-03	If a user initiates a Financial Ringi and the corresponding department budget is not available in the intermediary table, the system shall not allow the Ringi to be submitted. In such cases, the user may only save the Ringi as a draft until the required budget data becomes available. Once the required budget data becomes available in the system, the Ringi Creator shall then be allowed to submit the Ringi for further processing.
SIIT-04	The system shall use the new SAP budget codes received from the intermediary table for all budget-related processing.
SIIT-05	Budget consumption and budget reversal activities shall be synchronized with the intermediary tables in near real time to ensure accurate and up-to-date financial data.
SIIT-06	The new system shall not modify any SAP-side logic, data structures or perform direct create, update, or delete operations within SAP. All interactions with SAP shall be handled only through the intermediary tables as defined.
SIIT-07	Technology Services Provider shall be responsible for creating, maintaining and updating the intermediary tables required for the system. The synchronization between the intermediary tables and SAP shall be managed exclusively by Global Automotive Manufacturer and their SAP vendor.
SIIT-08	The system shall synchronize data with the intermediary table as follows: Write/Update Operations: Data created or updated in the new system shall be written to the intermediary table in real time. Read Operations: Data required from the intermediary table shall be fetched by the system at a scheduled interval of every 30 minutes.

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16. Document & Attachment Management

#	Requirement Description
DAM-01	The system shall support upload of multiple document types with configurable size and count limits by the Admin user.
DAM-02	System to support dynamic document types addition which can be configured by the Admin user. By default, the system shall support uploading of the following document formats for each Ringi: ● PDF ● DOC, DOCX ● XLS, XLSX ● CSV ● PPT, PPTX ● TXT ● RTF ● JPG, JPEG, PNG, TIFF ● XML, JSON ● ZIP ● Outlook mail file A maximum of 10 files can be uploaded per Ringi, including all files having a maximum size limit of 25 MB.
DAM-03	All uploaded files will undergo automatic virus and security scanning to ensure system safety. Hyperlinks in outlook mail to be disabled. Technology Services Provider shall assess the feasibility of disabling hyperlinks within attached files, including PDF, DOC/DOCX, and Excel formats. Implementation of this functionality shall be undertaken only if it is technically feasible.

17. Notifications & Email Alerts

#	Requirement Description
NA-01	The system shall provide automated, event-driven notifications to ensure timely action, transparency, and reduced manual follow-ups across the entire Ringi lifecycle.
NA-02	Notifications and Email shall be triggered automatically for the following events to Initiator: ● Ringi submission ● Approval at each approval level (L1, L2, L3, etc.) ● Rejection of a Ringi with mandatory remarks ● Ringi placed on Hold and subsequent Release ● Reassignment of approver or initiator ● Cancellation of a Ringi The system shall trigger notifications to an involved approver only when a Ringi is rejected by the immediate higher-level (+1) approver. In the case of approval, the system shall notify only the next higher-level (+1) approver and no notification shall be sent to any lower-level (-1) approvers. The Ringi

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	creator will be notified at every approval/rejection action.
NA-03	Notifications shall be sent to relevant stakeholders based on their role in the Ringi, including: ● Initiators ● Approvers ● Admins
NA-04	The system shall support email-based notifications using the mail server configuration provided by Global Automotive Manufacturer.
NA-05	Reminder notifications shall be automatically triggered for pending approvals based on configurable timelines defined by Global Automotive Manufacturer.
NA-06	Overdue approval notifications shall be escalated to Admin and the Ringi Creator to ensure timely decision-making.
NA-07	Upon final approval of a Ringi, the system shall automatically send the consolidated Ringi PDF to the initiator via email.
NA-08	Each notification shall include key contextual information such as: ● Ringi Number ● Ringi Type (Financial / Information) ● Ringi Amount (for Financial Ringis) ● Current approval level ● Pending action and responsible approver
NA-09	Notification templates and email content shall be approved based on inputs provided by Global Automotive Manufacturer.
NA-10	The system shall maintain logs of all notification triggers and deliveries for audit, tracking, and troubleshooting purposes.

18. Dashboards & Reporting

#	Requirement Description
DR-01	The system shall provide a centralized dashboard displaying all Ringis across the organization, including Financial, Information, Approved, Pending, Rejected, Cancelled, Hold Ringis accessible by Admin users only.
DR-02	The dashboard shall provide real-time visibility of Ringi status across plants, departments, and approval levels.
DR-03	The system shall highlight overdue Ringis that have been pending for action beyond a configurable threshold (e.g., more than 3 days).
DR-04	Role-based dashboards shall be provided as follows: ● Requester Dashboard: Displays all Ringis initiated by the user along with current status, approval level, remarks, and pending actions. ● Approver Dashboard: Displays Ringis pending for approval with priority indicators and aging

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	information. ● Admin Dashboard: Provides summarized metrics for the department, including approval trends, pending Ringis and budget utilization overview.
DR-05	The dashboard shall support dynamic search and filtering based on multiple parameters, including: ● Ringi Status (Pending, Approved, Rejected, Cancelled, Hold) ● Financial Year ● Department ● Plant / Location ● Ringi Type (Financial / Information) ● Capital / Revenue classification ● User Levels
DR-06	The system shall provide standard reports including, but not limited to: ● Total Ringis submitted, approved, rejected, pending, and reassigned ● Ringis by approval level and status ● Approval turnaround time by each Ringi lifecycle ● Overdue Ringis and bottleneck analysis by approver or department ● Budget utilization and reversal reports linked to Ringis - visible to all users. ● Turnaround time for a Ringi from the time of Ringi creation to the time of final approval The reports shall be exportable in excel format with Ringi details
DR-07	Reports shall be filterable using parameters such as: ● Date range ● Initiator ● Approver ● Approval level ● Budget range ● Financial Year ● Ringi Type ● Ringi Level
DR-08	Reports shall be exportable in multiple formats including Excel, CSV, PDF, and DOC.
DR-09	Access to dashboards and reports shall be controlled through role-based access permissions.
DR-11	The dashboard interface shall be user-friendly, plant-specific, and provide visual summaries to enable quick insights and management decision-making.
DR-12	The dashboard shall display reports only for the user's own cost centers.

19. User & Role Management

#	Requirement Description
URM-01	The system shall provide a centralized User & Role Management module accessible to Admin users.
URM-02	System shall be in-sync with existing HRMS to: ● Add new users ● Update user details

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	● Activate or deactivate users ● Assign or modify roles and permissions
URM-03	User onboarding shall be based on the HR master data provided by Global Automotive Manufacturer, including user attributes and organizational hierarchy.
URM-04	The system shall support synchronization of user data (new joiners, role changes, exits) as per the process defined by Global Automotive Manufacturer.
URM-05	The system shall support predefined standard roles: ● Creator ● Approver ● Admin
URM-06	The system shall enforce role-based access control (RBAC) to ensure users can access only the functionalities and data permitted to their assigned roles.
URM-07	In case of employee exit or role change, the system shall allow Admin to: ● Transfer ownership of pending Ringis ● Reassign approval responsibilities ● Maintain continuity of approval workflows
URM-08	All user management actions, including role assignment, access changes, and deactivations, shall be logged with user, timestamp, and action details for audit purposes.
URM-09	The system shall consume employee and organizational data from the existing HRMS as per the data structure provided by Global Automotive Manufacturer. The Employee Master data shall be synchronized with the HRMS on a daily basis.

20. Security & Access Control

#	Requirement Description
SAC-01	The system shall enforce secure user authentication using Single Sign-On (SSO) integrated with Global Automotive Manufacturer's existing identity management system.
SAC-02	The system shall restrict access to authenticated users only and prevent unauthorized access to application features and data.
SAC-03	Role-based access control (RBAC) shall be implemented to ensure users can access only those modules, actions, and data that are permitted based on their assigned roles.
SAC-04	The system shall support Microsoft Intune MDM integration to enable secure access for users connecting from outside the Global Automotive Manufacturer network, subject to Global Automotive Manufacturer's security policies.
SAC-05	The system shall enforce secure communication using HTTPS with TLS/SSL encryption.
SAC-06	The system shall use token-based authentication (JWT or equivalent) for maintaining secure user sessions.

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SAC-07	User sessions shall automatically expire after 30 days of inactivity.
SAC-08	The system shall prevent unauthorized actions such as approval, cancellation, or reassignment of Ringis by users without appropriate permissions.
SAC-09	The system shall maintain detailed audit logs for all security-relevant actions, including: ● User login and logout ● Role and permission changes ● Approval, rejection, cancellation, and reassignment actions ● Configuration changes by Admin / Super Admin users
SAC-10	Audit logs shall capture the user ID, action performed, date/time stamp, and relevant reference details.
SAC-11	The system shall restrict access to sensitive reports and administrative configurations based on role permissions.
SAC-12	Uploaded documents and system data shall be protected from unauthorized access based on role and Ringi-level permissions.
SAC-13	The system shall comply with Global Automotive Manufacturer's internal IT security and audit guidelines.
SAC-14	Users logging in via SSO shall not require OTP validation, whereas users logging in using credentials shall be validated through email-based OTP.

21. User Manual Documentation

#	Requirement Description
UMD-01	Technology Services Provider shall provide comprehensive user manuals for the application, covering usage guidelines for Ringi Creators, Approvers and Admin users. User Manuals to be in Japanese also.
SAC-02	The user manual shall be made available in PDF format and accessible directly through the application portal for all authorized users.
SAC-03	The user manual shall include step-by-step instructions with relevant screenshots to guide users in performing actions within the system.

22. Language Support (English to Japanese)

#	Requirement Description
LS-01	The system shall support multi-language capability with primary support for English and Japanese.
LS-02	Japanese language conversion shall be applied only to the Ringi body content, all other pages shall

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	support English conversion only. System data shall be stored exclusively in English. Japanese language conversion shall be performed at runtime for the Ringi body content and shall not be persisted in the database.
LS-03	All amounts mentioned in Ringis and budgets shall be displayed in Indian Rupees (INR) only.
LS-04	Technology Services Provider shall use the Microsoft Translator Text API (Azure) for translation functionality within the system. The subscription and associated costs for the translation service shall be borne by Global Automotive Manufacturer.

23. Login & Authentication

#	Requirement Description
LA-01	The system shall support Single Sign-On (SSO) authentication using Global Automotive Manufacturer's existing enterprise identity management solution.
LA-02	Users shall be authenticated using their corporate credentials, eliminating the need for separate usernames or passwords for the Ringi application.
LA-03	The system shall rely on the organization's centralized authentication mechanism (e.g., Active Directory / Azure AD) for validating user identity.
LA-04	Upon successful authentication, users shall be automatically logged into the Ringi system and granted access based on their assigned roles and permissions.
LA-05	The system shall not store or manage user passwords locally; all authentication shall be handled through the configured SSO mechanism.
LA-06	User profile details such as name, employee ID, department, and designation shall be fetched from the identity provider or user master data maintained by Global Automotive Manufacturer.
LA-07	The system shall validate user authorization post-login to determine: ● Accessible modules ● Permitted actions ● Applicable approval levels
LA-08	All login and logout activities shall be logged for audit and security monitoring purposes.
LA-09	The login mechanism shall be compatible with Global Automotive Manufacturer's existing IT security policies and infrastructure, including integration with Microsoft Intune where applicable.

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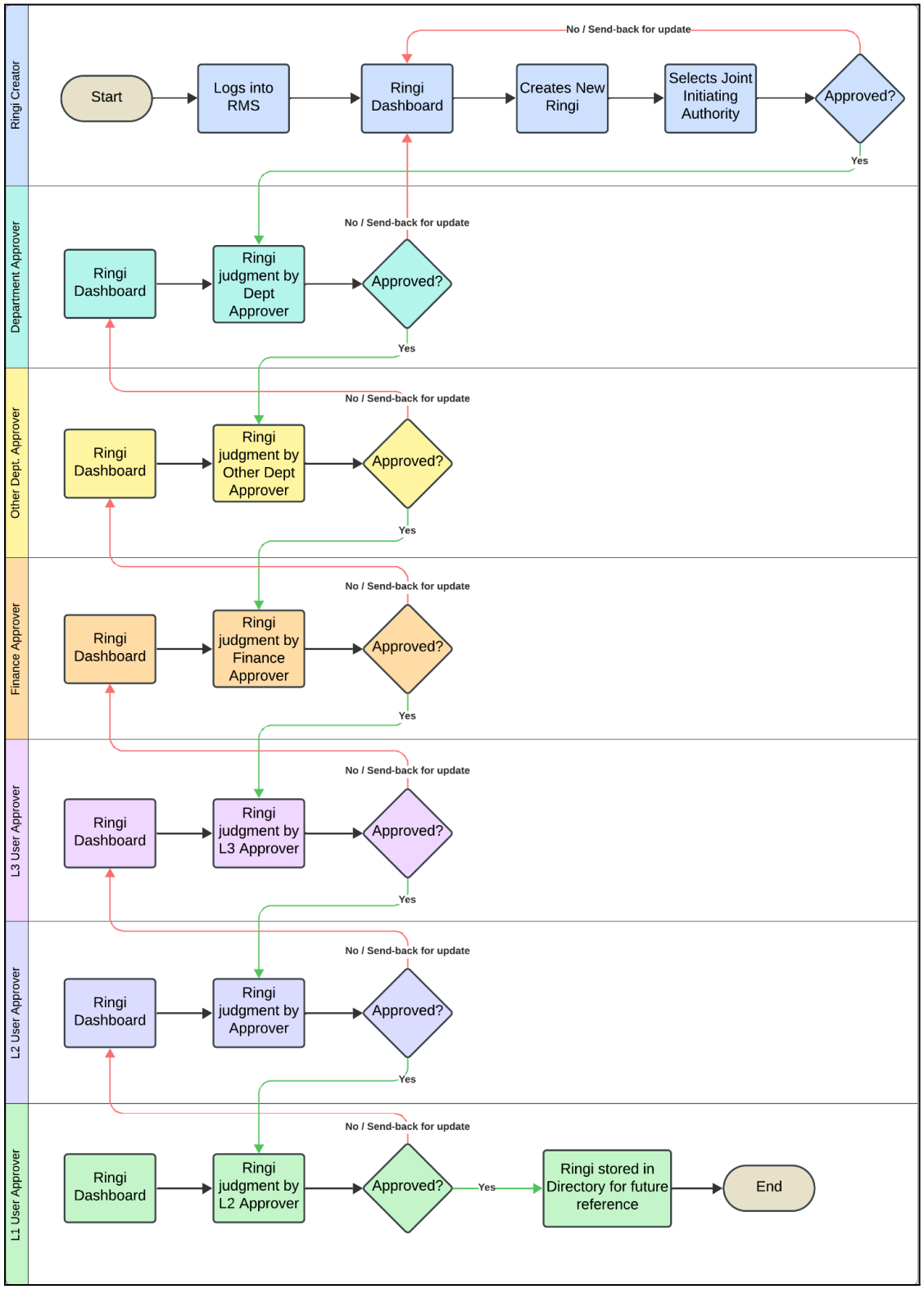
Non-Functional Requirements

ID	Requirement
NFR-001	Data should be secured with role-based access control (RBAC).
NFR-002	Logs for registration, approval, login, and password reset to be maintained.
NFR-003	Should handle a growing number of Ringi requests.
NFR-004	Every change must be tracked with timestamp and user details.
NFR-005	Images uploaded to the system must use lossless compression to optimize storage space.
NFR-006	The system shall support concurrent access by multiple users without performance degradation.
NFR-007	Background processes such as notifications, audit logging and intermediary table synchronization shall not impact user-facing performance.
NFR-008	The system shall comply with Global Automotive Manufacturer’s internal IT security policies.
NFR-009	Audit logs shall be immutable and accessible only to authorized personnel.
NFR-010	The application shall have an intuitive, user-friendly interface requiring minimal training.
NFR-011	The system shall be accessible via standard web browsers on desktop and laptop devices.

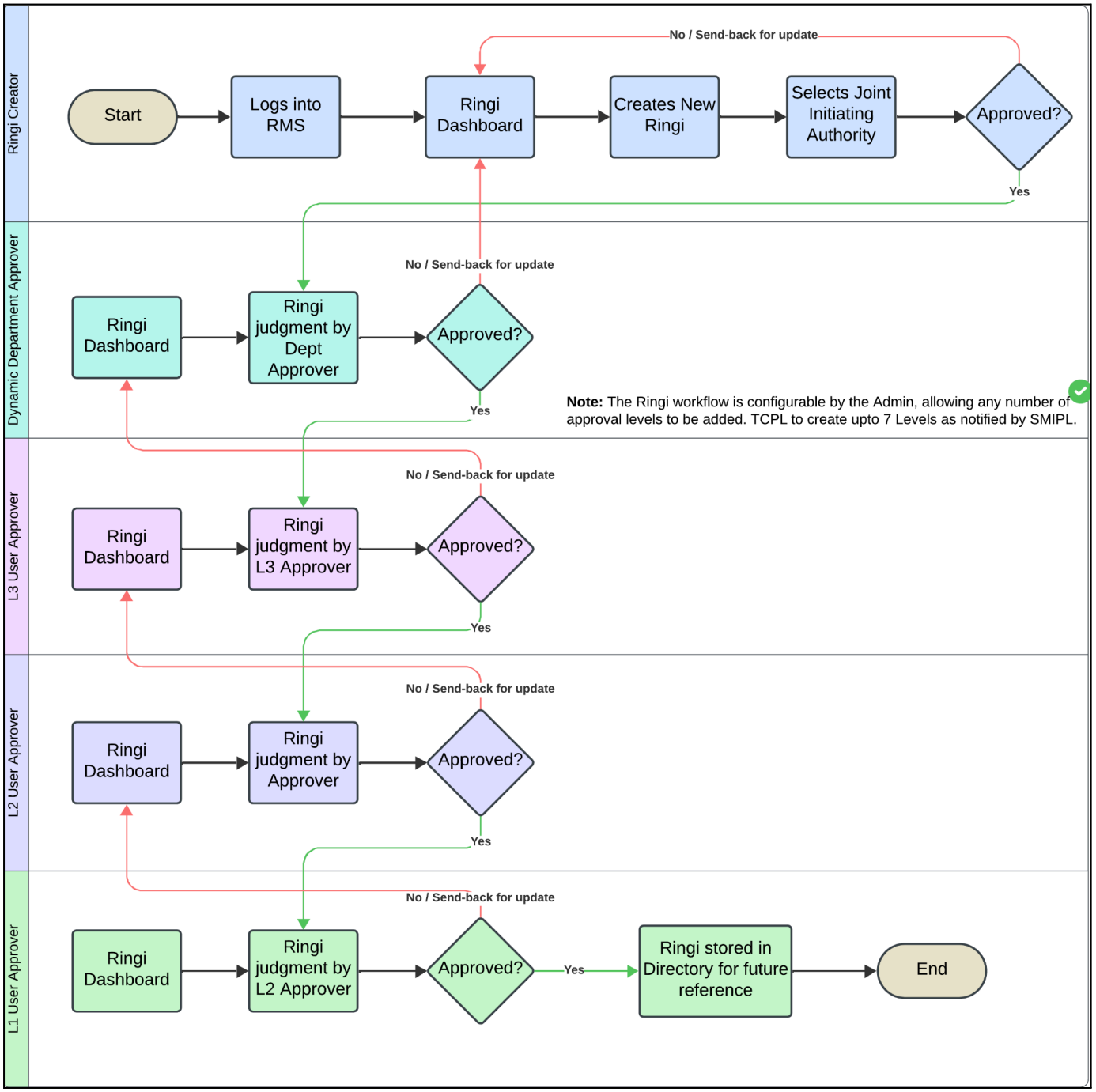
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Process Flow

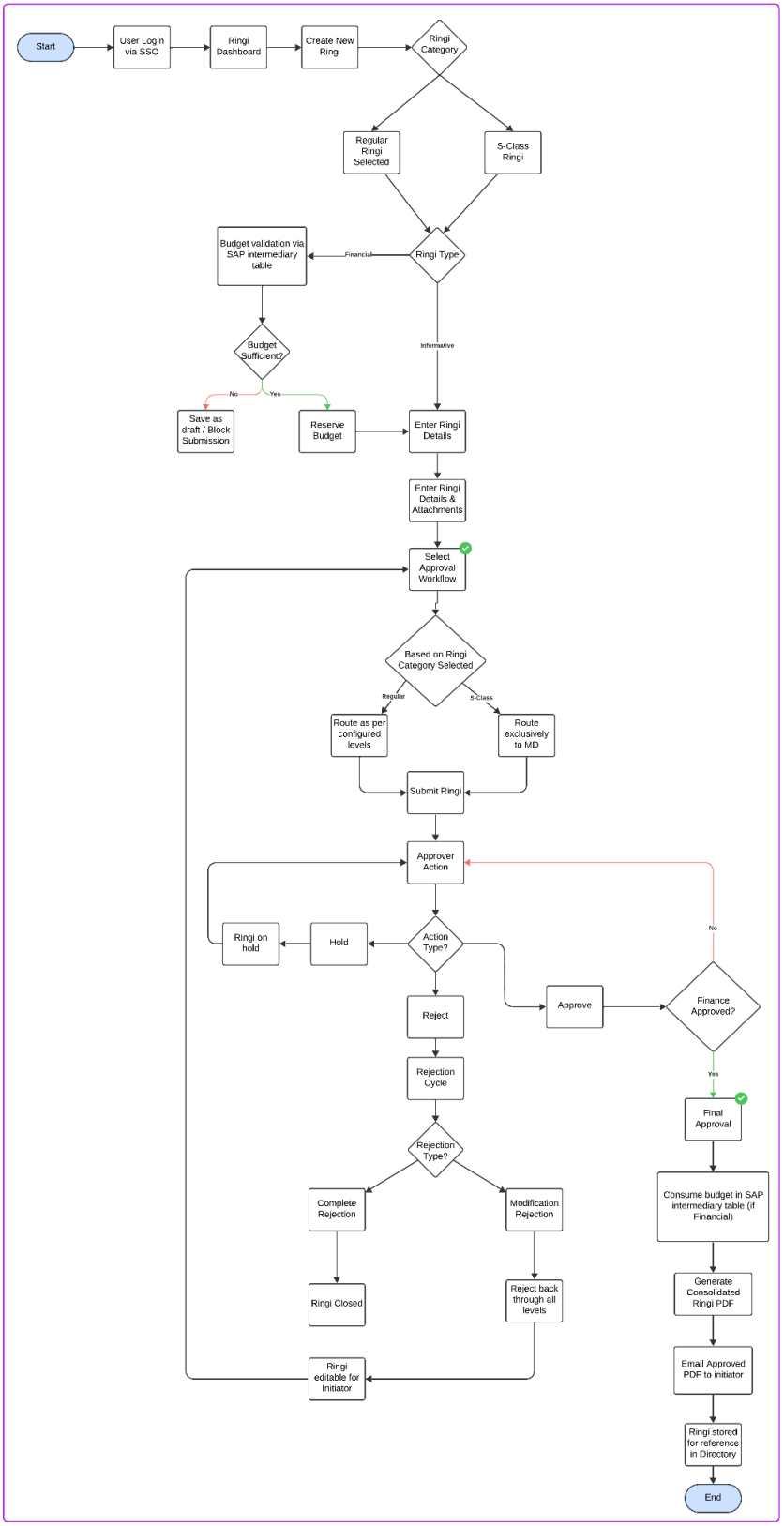
- Current Process Workflow



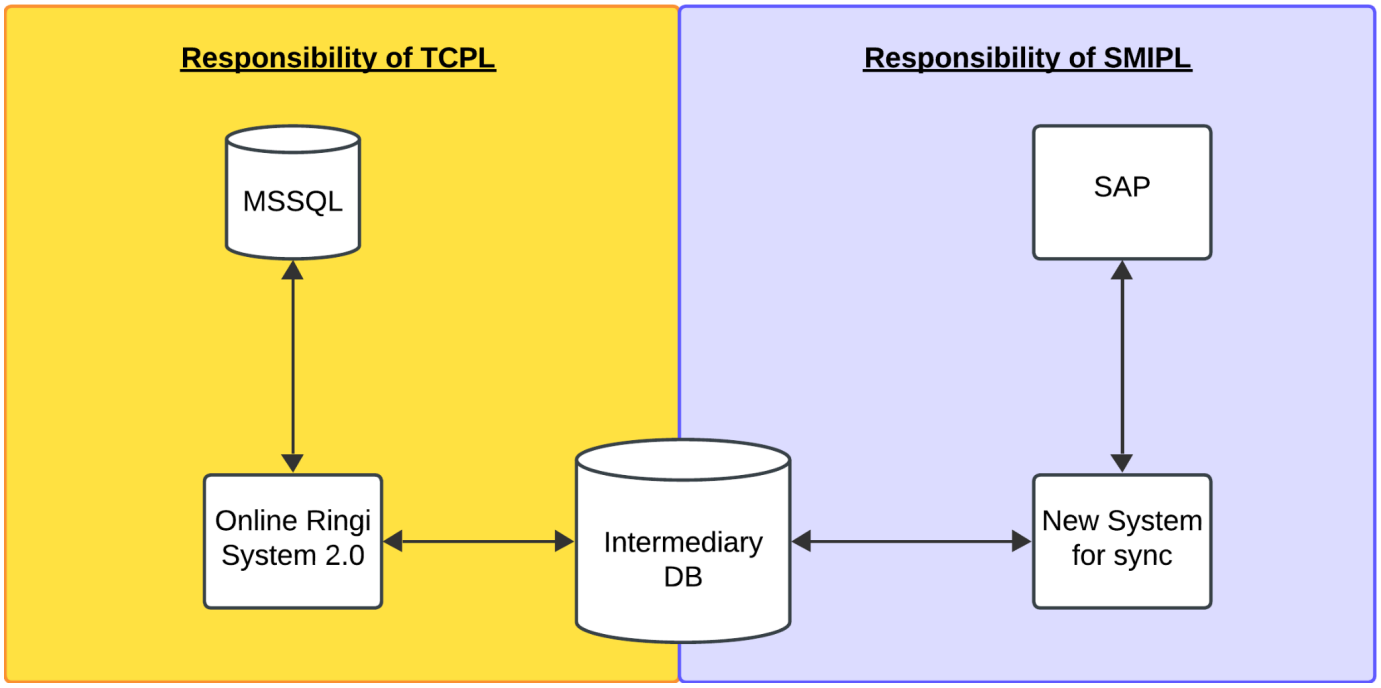
- To-Be Process Workflow



- Ringi Creation Flow



- Data Ownership and System Responsibility Overview



Hardware and Software Requirements

The ORS will be designed to support up-to 350 concurrent users, document storage growth of 30 GB per year, and database growth of 10% annually over a 10-year lifespan.

Technology Platform Recommended by Technology Services Provider:

Platform	Technology
Backend Development	ASP .NET Core
Frontend Development	Angular
Database	Microsoft SQL Server 2022 Standard/Enterprise
UI	HTML, CSS, JavaScript, Bootstrap
Hosting	On Prem
Security	HTTPS, JWT-based authentication Role-based access control Encrypted document storage
Integration	HRMS

1. **Application Server (Angular + ASP.NET Core)**

- CPU: 4-core
- RAM: 16 GB DDR4 ECC
- Storage: 100 GB SSD (OS, Angular builds, logs)
- OS: Windows Server 2022
- Deployment: IIS
- Backup : Daily Incremental + Weekly Full

2. **Database Server (SQL Server)**

- CPU: 4-core high-frequency
- RAM: 24 GB ECC
- Storage: 150 GB SSD (50 GB OS, 20 GB logs, 50 GB DB, 30 GB backups)
- SQL Edition: SQL Server 2019/2022 Standard or Enterprise
- OS: Windows Server 2022 Standard
- Backup : Daily Incremental + Weekly Full

3. **Document/File Storage**

- Storage: 150 GB SSD/NAS (130 GB Production+ 20 GB QA)
- Backup: Weekly snapshots
- Backup : Daily Incremental + Weekly Full

4. **QA/Dev Server (Angular + ASP.NET Core +SQL Express Edition)**

- CPU: 4-core
- RAM: 24 GB DDR4 ECC
- Storage: 200 GB SSD (OS, Angular builds, logs, SQL Express)
- OS: Windows Server 2022
- Deployment: IIS
- DB Engine : SQL Server 2022 Express Edition
- Backup : Daily Incremental + Weekly Full

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5. Software & Licensing

- Windows Server: 2x Standard licenses
- SQL Server: 2019/2022 Standard or Enterprise
- Antivirus: Any

6. Additional Details

- Server Availability : 24*7
- Public IP
- SSL Certificate
- Antivirus
- SMTP

7. Final Notes

- The system must handle up to 350 concurrent users efficiently.
- Annual performance and hardware reviews are recommended.
- Long-term success requires proper backup and scalability planning.

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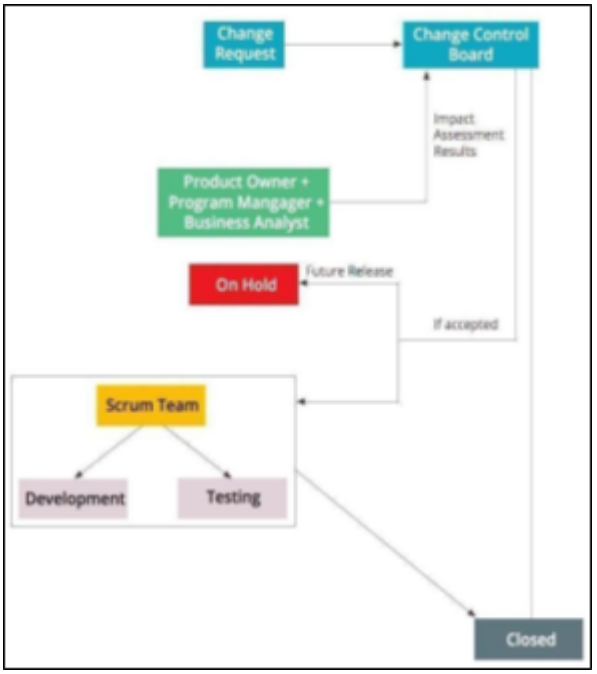
Change Request (CR) Management

1. Change Request (CR)

Technology Services Provider proposes that changes to requirements be handled using the project change request management tools and procedures duly signed off by the client and the Technology Services Provider project-in-charge. The major stakeholders of the project need to have a clear understanding of these changes and the associated implications on the project effort, timelines, cost, and related systems. Technology Services Provider, along with client project managers and business stakeholders, will work effectively to absorb variations from initial functional requirements and discuss trade-offs to balance scope, schedule, and resources (budget), if possible.

If Global Automotive Manufacturer wishes to introduce different goals or principles based upon new/added factors introduced or discovered by the teams, including, but not limited to, the type of work to be done, the number of hours of work to be performed, the schedule estimated, or a change in management. Direction affecting the current scope of the project, or any other changes in direction, priority, schedule, or approach once established and agreed to in this PROPOSAL, must undergo this CR process. Such changes will necessarily have both a schedule and cost impact on the project, and these changes will not be binding upon Technology Services Provider within the current scope and shall have both an equitable price and schedule adjustment. Any such change requests duly signed off will be executed by Technology Services Provider on an additional time, cost, and effort basis.

2. CR Management Approach



The goal of the change management process is to produce a more robust system that better supports clients' business activities.
The change management process includes—

- **Collect** change requests (a generic term encompassing defects, bugs, requests, and suggestions). This is a continuous process and will be done during each sprint sign-off.
- **Identify** the scope of the next release and determine which change requests will be included in the next build. The change requests should be grouped according to the functional and/or physical area of the system the request will change.
- **Document** the requirements, functional specifications, and implementation plans for each grouping of change requests within the Product/Project with a proper sign-off
- **Implement the change.** This may involve changes to source code, database structure, or user documentation.

- **The test** is to verify the changes done by the person who made the change, usually the programmer. It is very specific to just the area changed. Tests a functional area of the system to see that everything works as expected.
- **Release** the system to Product/Projection. This is the physical process of upgrading the product/projection system to the new release. This may involve executable files, database changes, data migration, and the installation of new support software.
  Change management impact could be a variance to the budgeted efforts, timelines, and cost of the original project scope of work.

3. Definition of Change Request

The following will be categorized and treated as change requests.

1	Any specific change or enhancement requested beyond what is explicitly defined and agreed upon in the approved Business Requirements Document (BRD).
2	Delays in providing formal sign-off on the BRD after its submission, where the sign-off timeline varies based on ongoing discussions, clarifications, or mutual communication between the involved stakeholders. The sign-off timeline shall be subject to change based on agreed communication and coordination.
3	Delays in providing formal User Acceptance Testing (UAT) sign-off after completion of UAT activities, where the sign-off timeline is influenced by feedback cycles, discussions, or business considerations. The sign-off timeline shall be subject to change based on agreed communication and coordination.
4	Any delays caused due to the unavailability, incompleteness, or delayed provision of required data, where such unavailability results in a pause or impact on project activities. The duration and impact of such delays shall be subject to discussion and mutual communication between the stakeholders.
5	Any delays arising from issues related to the availability, access, or stability of infrastructure and/or the customer's internal core systems required for development, testing, or deployment activities. The handling of such delays and associated timelines shall be subject to discussion and mutual communication between the stakeholders.

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Dependencies, Assumptions, Out of Current Scope

A. Dependencies on the Global Automotive Manufacturer

- The Global Automotive Manufacturer is expected to provide the private & public domain URL, SSL certificate, server access for development and production environments, and branding guidelines (logos, color schemes, fonts, and other branding elements).
- Timely availability of business stakeholders for requirement validation, walkthroughs, and approvals.
- Final sign-off on BRD, workflows, and configuration rules before development commencement.
- Timely feedback and sign-off on UAT findings to avoid project delays.
- Infrastructure for development/sandbox and production to be provided by the Global Automotive Manufacturer team.
- All master data is to be provided on time and as per the finalized/Technology Services Provider template. All the portal content/images/banners/policy documents are to be provided by the Global Automotive Manufacturer team.
- All meetings should happen on time without any delay.
- All the corporate branding guidelines are to be provided by the Global Automotive Manufacturer team for the UI/UX development.
- The Global Automotive Manufacturer team needs to ensure the availability of users at the time of testing and training.
- Any deliverables (e.g., approvals, UATs, sign-offs, etc.) submitted to the Global Automotive Manufacturer team should be reviewed and signed off within 5 working days of delivery (since this is a tightly time-bound project).
- Availability of business users for User Acceptance Testing (UAT) as per the agreed schedule.
- Any approval provided on email will be considered as formal approval and sign-off.

B. Assumptions

- Business stakeholders from Global Automotive Manufacturer will be available for requirement discussions, validations, and approvals as per the agreed project plan.
- All business requirements will be finalized and signed off during the BRD phase.
- Any change requested after BRD sign-off will be treated as a Change Request (CR).
- Timely inputs and decisions will be provided by designated business and IT SPOCs.
- The application will be hosted on infrastructure provided and managed by Global Automotive Manufacturer.
- Required environments (Development, Production) will be provisioned and accessible as per project timelines.
- SAP systems and APIs required for budget validation and data exchange will be available, stable, and accessible during development and testing.
- User master data, roles, and hierarchy will be provided and maintained by Global Automotive Manufacturer.
- UAT will be conducted by designated Global Automotive Manufacturer users within agreed timelines.
- Test cases and test data will be provided or validated by the business team.
- UAT sign-off will be provided formally upon successful completion.
- Delays in UAT execution or feedback may impact overall project timelines.

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- Any changes to user access, roles, or organizational structure will be communicated in advance.
- The system will rely on Global Automotive Manufacturer's identity management mechanism (SSO) for authentication.
- All integration specifications, credentials, and endpoints will be shared by Global Automotive Manufacturer in a timely manner.
- No changes will be made to SAP-side logic or data structures as part of this project.
- Only master data (users, departments, cost centers, budget codes) will be shared for system setup; historical Ringi data migration is not in scope unless explicitly agreed.
- Network connectivity, firewall rules, and security configurations will be made available to enable system access and integrations.
- SSL certificates, domain configuration, and required ports will be provided by Global Automotive Manufacturer.
- The Global Automotive Manufacturer team has a solid understanding of their own business in teORS of processes and system requirements. All our project understanding and execution are based on the inputs shared by the Global Automotive Manufacturer team. Hence, post go-live, if errors are uncovered about the understanding previously given, thereby requiring Technology Services Provider to reset up or re-implement (even if partially), the same would be executed on an additional cost, time & effort basis.
- Any repeat work undertaken by Technology Services Provider due to miss/failure by the Global Automotive Manufacturer team will be executed on an additional cost, time & effort basis.
- All UAT, training, and Conference Room Pilot (CRP) activities are considered as one-time-only activities to optimize project timelines. If multiple iterations of the aforesaid executions would be required, the same would impact the project execution timelines and cost, which will be treated as a Change Request (CR) project.
- All UATs are restricted to only the deemed users and not any other observer of the project.
- The portal will be accessible to the Global Automotive Manufacturer members only. It depends on the Global Automotive Manufacturer team to decide the authentication and authorization of the portal.
- The Global Automotive Manufacturer IT team will be the SPOC for this project and will communicate with the business team.
- Hypercare will be initiated after the post-live, and after the expiry of the hypercare, the support contract needs to be signed for the support on an additional cost, time & effort basis.
- User authorization data will be provided by the Global Automotive Manufacturer team, and it will be maintained (addition/modification) by the Global Automotive Manufacturer team.
- Post-go-live portal admin activity is to be performed by the Global Automotive Manufacturer team.
- There will be no option to change the role-wise authority in case the respective authority is not available.
- All the development will be done offsite from the Technology Services Provider office locations.
- Any data changes due to version compatibility will be maintained by the Global Automotive Manufacturer team.
- All the Repair & Quality Inspection Portal content/images are to be provided by the Global Automotive Manufacturer team.
- The social media access control and content display over the Repair & Quality Inspection Portal links are to be provided by the Global Automotive Manufacturer team.

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- Project Delivery Completion Document, which will be required to be signed off by the Global Automotive Manufacturer along with the Go-Live sign-off.

C. Out of Current Scope

- While creating a new Ringi, only a single budget code can be linked. Linking multiple budget codes to a single Ringi shall not be supported.
- Development, modification, or enhancement of SAP modules, configurations, or business logic.
- Creation or modification of SAP reports, transactions, or workflows.
- Integration with any systems other than those explicitly defined (e.g., HRMS, Payroll, Vendor Portals, third-party ERP systems).
- Migration of historical Ringi data, approvals, or documents from legacy systems.
- Migration of historical budget, financial, or audit data.
- Validation or reconciliation of legacy data.
- Procurement, configuration, or maintenance of servers, databases, or network hardware.
- Support for offline or desktop-based application usage.
- Additional integrations not listed in the approved scope.
- Native mobile applications (Android / iOS) are not included.
- Only browser-based access via responsive web interface is covered.
- Setup or maintenance of operating systems, firewalls, VPNs, or security appliances.
- Performance tuning at infrastructure or database level beyond application optimization.
- Any data correction, cleansing, or reconciliation within SAP.
- Historical data migration from previous systems, spreadsheets, or paper-based check sheets is excluded.
- No separate mobile app will be developed for inspection or reporting. The application will be browser-based, accessible via tablet or desktop.
- Any new process, new functionality, new application, new workflows, etc. not mentioned in the scope. The same would be catered through the Change Request (CR) process on a time, material, and cost basis if this new scope is relevant to this project scope. If the new/added scope is directly relevant to the current scope and requires a fresh development, the same might be executed as a new project phase, based on a mutual agreement on the same.
- Cost of any other software, infra., tools, etc. required for implementation and deployment of the application
- Any issues arising due to infrastructure and/or reasons not in control of Technology Services Provider.
- No third-party tool will be implemented/integrated into the portal unless otherwise already mentioned in this scope document.
- Management of hardware of the database and applications.

Project Communication and Control

A. Documentation

The project review call/meeting is planned every 7 working days. The mode of the review can be direct or web, whichever is agreed upon between the project managers of both divisions. The Project Status Report (PSR) is sent to the stakeholders after the meeting conclusion.

The Technology Services Provider team will give the following documents to the Global Automotive Manufacturer team during the project execution.

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- Design Prototype
- User manual document
- Weekly Project Status Report.

B. Governance

- Governance will be done leveraging a project tracker and a periodic project execution review, which will be attended by the project SPOCs and relevant team members of both Global Automotive Manufacturer and Technology Services Provider.
- A collaborative governance model for smooth IT operations, issue resolution, and escalation process for effective monitoring and quality control of project execution is followed as a standard practice by Technology Services Provider.
- This governance will lay an effective relationship management with an evolution of the account governance framework, as our relationship grows from a service provider to a trusted advisor.
- Such governance can also help in proactive forecasting of clients' upcoming solution & professional resource needs by gauging business requirements, operations & supporting domain functions.
- The general practice followed by Technology Services Provider to facilitate all of the above is as depicted below.

Governance Category	Frequency	Governance Members
PSR (Project Status Report)	Fortnightly	Project Task Owners / Project SPOCs
Project Track Meetings	Once every month	Project Managers & Delivery Head
Leadership Sync	Once every Quarter	Project Sponsors, Business Heads

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